

NEWPORT HARBOR, RI, USA

WMO: 997285

Lat: 41.504N

Lon: 71.326W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.8	(c) -8.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.0	(k) 2.8	(l) 12.4	(m) 2.4	(n) 6.5	(o) 0	(p) 0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 4.9	(c) 27.3	(d) N/A	(e) 25.8	(f) N/A	(g) 24.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.3	(p) 340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.0	(b) 10.5	(c) 9.0	(e) -12.7	(f) 31.5	(g) 2.7	(h) 2.1	(i) -14.6	(j) 33.0	(k) -16.2	(l) 34.2	(m) -17.7	(n) 35.3	(o) -19.6	(p) 36.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-0.1	0.4	3.5	8.8	13.4	18.4	22.4	21.9	18.8	13.9	8.4	3.5
	DBStd	8.54	4.95	4.04	3.93	2.85	2.87	2.71	1.90	1.83	2.48	3.33	3.99	4.23
	HDD10.0	1132	313	270	203	54	3	0	0	0	0	7	77	204
	HDD18.3	2933	571	503	460	285	156	31	1	1	25	141	299	461
	CDD10.0	1556	0	0	1	19	109	253	383	369	263	128	27	2
	CDD18.3	315	0	0	0	0	3	34	125	112	38	4	0	0
	CDH23.3	688	0	0	0	0	10	60	327	245	41	5	0	0
	CDH26.7	71	0	0	0	0	2	7	39	21	1	1	0	0
Wind	WSAvg	4.4	4.9	4.9	5.1	4.8	4.3	3.9	3.7	3.4	3.7	4.5	4.8	4.9
Precipitation	PrecAvg	1176	93	88	120	105	88	91	84	96	102	99	99	111
	PrecMax	1425	157	165	387	204	188	271	257	199	235	330	171	183
	PrecMin	942	42	45	7	39	29	0	25	20	21	9	19	32
	PrecStd	135	31	34	77	43	34	67	45	46	49	66	40	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	10.1	14.4	20.1	24.5	27.3	30.1	28.8	26.3	23.3	17.5	14.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	10.6	8.2	11.7	15.8	20.8	25.0	27.6	26.9	24.6	21.0	16.2	12.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	8.9	6.9	9.9	14.1	19.0	23.3	26.2	25.5	23.2	19.6	15.2	11.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	7.2	5.8	8.6	12.7	17.4	22.0	25.1	24.7	22.3	18.7	14.4	10.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.0	6.0	5.9	5.6	5.3	4.9	4.9	5.2	5.3	5.6	5.8
		MCDBR	7.6	6.6	7.7	8.2	8.0	7.3	6.7	6.0	6.0	5.6	5.8	7.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.307	0.315	0.338	0.375	0.418	0.462	0.487	0.444	0.377	0.346	0.317	0.305
	taud		2.504	2.481	2.455	2.385	2.299	2.214	2.152	2.265	2.428	2.519	2.558	2.557
	Ebn at Noon		851	898	912	896	861	820	794	819	859	851	834	825
	Edh at Noon		73	87	100	115	129	141	149	129	102	83	70	65
All-Sky Solar Radiation	RadAvg		1.87	2.75	3.77	4.86	5.38	5.89	6.04	5.42	4.42	3.00	2.04	1.57
	RadStd		0.15	0.27	0.27	0.52	0.52	0.58	0.32	0.28	0.32	0.27	0.14	0.15

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+96	+47

Nomenclature: See separate page